FedCast

A forecaster for the next Federal Reserve rate decision, built to be judged by its process rather than its luck

Tan Kah Ming · Fullerton · October 2026

About me

Tan Kah Ming

Senior AI, Data and Technology Consultant, seconded to Temasek Holdings, Digital Innovation (2023 to now)

Before that

- Data scientist, RSAF Data Office: authored the Air Force's AI strategy, set up its first digital unit
- Air engineering and weapons logistics officer, where the data-driven habit started (Weibull analysis, predictive maintenance)

Training

M.Eng Aeronautical Engineering, Imperial College London · Graduate Certificate in Predictive Analytics, NUS · CFA Level 3 candidate (February 2027)

Things I have built that models had to earn trust in

1

Investment analytics at Temasek

An AI-assisted analyst performance and attribution platform for 50+ investment professionals, with an LLM signal tracker for high-conviction ideas; signal cadence moved from monthly to daily, every recommendation logged in real time.

2

Valuation and execution

A machine-learning mark-to-market for private holdings, systematic and reproducible; and an AI-driven order management system on AWS processing about S$30m a day, replacing an email-based workflow.

3

Forecasting under a scoreboard

4th of 160+ teams (MIT, DoD, USAF) in the 2023 USAF AI Accelerator, predicting pilot flying difficulty from physiology; an ammunition recommender (quantile LightGBM) that cut wastage with no shortfall, Defence Manpower Committee Innovation Award 2022.

4

COVID-19, 2021

Built, in weeks, the system that engaged every home-recovery patient in Singapore, 10,000 people a day, on form.sg, Azure and Zendesk. The lesson that carries over: a system people rely on must say what it knows and what it does not.

The question, and today's answer

28 Oct 2026

the decision being forecast

71%

no change

23%

+25 bps hike

6%

the other three outcomes together

Where the number comes from

- One frozen snapshot of 22 evidence items, pulled 26 October: futures, eight FRED series, both statements, minutes, transcript, projections, speeches, 32 years of decisions, prediction-market odds
- Four models pooled by code with fixed weights: market-implied 0.6, the Fed's own Taylor rules 0.2, prediction-market odds 0.1, historical base rates 0.1; then a 1.5% floor on every outcome
- Five specialist analysts (a language model, tools off, every claim quoted and verified) lean the pooled number by a bounded 4.4 points toward a hike
- Ledger entry #16, hash-chained, replays exactly from its snapshot

Live demo 1: the forecast

1 min

Open

- The app at localhost:8765, Forecast view
- Top bar: when the evidence was pulled, what is new at the sources

Point at

- The two tracks side by side: machine only, and with views
- The meta line: ledger entry, snapshot hash, the specialists' tilt
- The five bars

Say

- This number was made at <time> from evidence frozen at <time>. Everything I show next is how it got there.

LIVE DEMO

Why build this

1

Judge the process, not the luck

Eight decisions a year is far too few to tell skill from chance. So the yardstick is a ten-dimension scorecard fixed before any code existed: repeatability, groundedness, leakage, directional sanity, noise invariance, coherence, auditability and more. Outcomes are logged and never used to tune.

2

A number you can audit

Every forecast runs on one fingerprinted snapshot. Every model lists the items it read. Every analyst claim carries a verbatim quote that code has matched. Every entry in the ledger recomputes exactly from its snapshot. Nothing moves the number silently.

3

A harness that transfers

The decision is the first target, not the only one. The same chain can forecast the December meeting, the year-end rate, or another central bank, and the control analysts measure whether the language model is still earning its place.

What others do, and what each lacks

Predictor	What it is	Strength	What it lacks
Fed funds futures (CME FedWatch)	Price of the contract that settles on the month's average rate	About 88% right at 30 days over 1994 to 2024 (Bonini, Huang, Simaan). The anchor of our pool	No reasons. One number with no evidence trail; fails silently when the Fed surprises
Prediction markets (Polymarket, Kalshi)	Crowd odds on each outcome	Independent crowd; fast to move	Thin and fee-distorted; a $22 entry can sit next to a $67m one
Economist surveys	Median of polled forecasters	Human judgement, broad	Slow, unauditable, consensus-seeking
Policy rules (Taylor family)	Rate a formula says is appropriate	Transparent, the Fed's own benchmarks	Depends on unobservable r* and u*; says where rates should be, not what the committee will do
LLM 'FOMC agent' papers (2025)	Models that read Fed text and vote	Reported 90%+ accuracy on 2023 to 2024	16 meetings; no frozen evidence, no quote verification, no replay, no control
FedCast	All of the above pooled by code, plus verified analysts	Every number traceable; repeatable; a scorecard that catches drift	No live record yet; no accuracy claim at five weeks

Sources: Bonini, Huang and Simaan, 'Watching the FedWatch' (232 meetings, May 1994 to March 2024); FedSight AI preprint, December 2025; Takano, Hirano and Nakagawa, arXiv 2511.02469.

Our edge: not a better guess, a checkable one

Frozen evidence

One fingerprinted snapshot per forecast, dated in US time, data as first published. No internet, no memory. If it is not in the snapshot, nothing may read it.

Code computes, models read, code checks

Arithmetic is done by code. The language model is used only for reading text, and every claim it makes must quote a snapshot item; code string-matches the quote and flags any number not in the evidence.

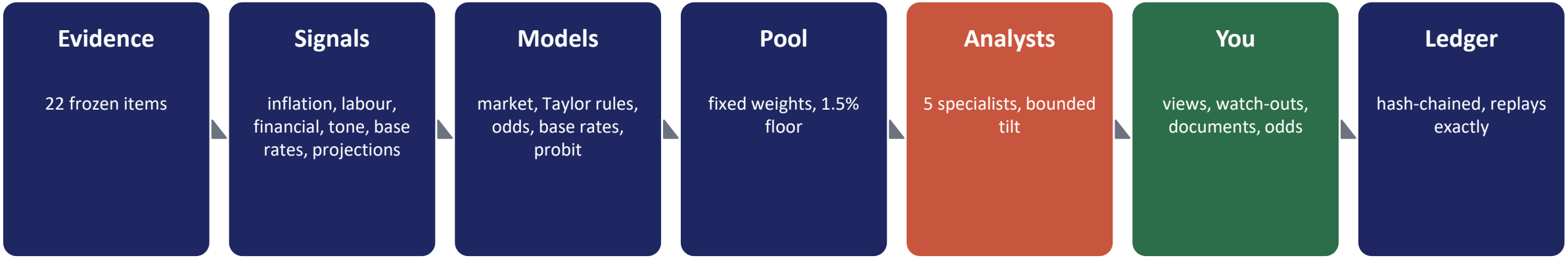
Human judgement through typed channels

Views tilt the number by a capped amount and expire. Watch-outs must be answered by id. Documents and odds enter as evidence. Nothing nudges the number quietly.

Pre-registered, amended in the open, replayable

The scorecard, weights, floor and every model rule are hashed and pinned in the tests. Seven logged amendments. Every ledger entry recomputes from its snapshot.

One chain, left to right



Signals are computed, not asked: the analysts are handed the numbers code has already worked out. Everything a model could compute, code computes. The language model reads the words.

Today's readings: from five models to one number

Model	What it reads	−25	no change	+25	+50	weight
Market-implied	November fed-funds future against the overnight rate: +2.5 bp expected	0	90	10	0	0.6
Taylor family	12-month core PCE 3.0%, unemployment 4.2%, four rules × nine r*/u* pairs	9	44	40	7	0.2
Prediction market	Polymarket, 26 Oct, $67m traded	0	95	4.5	0	0.1
Historical base rates	What followed a hike at the 49 meetings since 1994 that came after one	0	35	52	13	0.1
Pooled	Weighted average, then the 1.5% floor	**1.8**	**74.7**	**19.4**	**2.6**	
Specialists' tilt	+4.4 bp hawkish, mean of five: 4.4 points moved toward the hike outcomes					
Machine forecast	Ledger entry #16	**1.4**	**71.3**	**22.6**	**3.8**	

Percentages, rounded. The ordered probit is computed and shown but carries weight 0 until it has a live record.

Live demo 2: the evidence graph and the pool

3 min

Open

- Evidence graph
- Click the Market-implied box
- Click the Pool box
- Methodology, scroll to the worked example

Point at

- Market-implied: the futures price, 100 minus price, the expected change, the 90/10 split
- Pool: the four weights, the pooled line, the specialists' tilt line, machine_only
- Any evidence item: the frozen text and the link to the original

Say

- Blue is in the number, grey is computed but not in it. Click a model and you see only what it read.
- The language model's lean is the one line between pooled and machine_only. It can tilt, it cannot write.

LIVE DEMO

Eight analysts, one model, one ladder

1	**Analyst 1: naive** The question and the date. Nothing else.	What the model does on its own. The control every later step is measured against.
2	**Analyst 2: evidence shown, nothing verified** Every document and every pre-computed number, 128,000 characters. Free-form answer.	Adds the evidence. Shows what evidence alone fixes, and what it does not.
3	**Analyst 3: generalist with verifier** Same evidence, but a fixed answer shape: a verbatim quote per claim, a watch-out report, a tilt capped at ±10 bp. Code checks every quote and number.	Adds structure and verification. This is where the output becomes stable and checkable.
4-8	**Analysts 4 to 8: specialists** One slice each: communications, minutes, inflation, labour, financial conditions. Same schema, same verifier.	Adds specialisation. Each judgement is attributable to its evidence; a dissent survives instead of being averaged away.

Each step adds one thing. The value of that thing is the difference in score between neighbouring steps.

What the harness enforces around every analyst

1. **Tools off, slice only**

   The model sees its slice of the frozen snapshot and nothing else: no browsing, no memory of the Fed, no other analyst's output.

2. **Pre-reads**

   Numbers are computed by code and handed over: inflation gap, labour slack, financial conditions, tone, base rates. The model never does arithmetic.

3. **Fixed answer shape**

   Claims, each with a verbatim quote and the item it came from; a report on every watch-out by id; a tilt bounded to ±10 basis points. Enforced by the runtime.

4. **Code verifies**

   Every quote is string-matched to its item. Every number is matched to the snapshot. Every watch-out must be answered. Failures are counted, never hidden.

5. **Recorded and repeated**

   Five identical runs per analyst per snapshot, each stored with its prompt hash. Repeatability is measured, and a recorded run replays for free.

Live demo 3: what each analyst is asked and given

2 min

Open

- Analyst prompts view
- Analyst 1: open the full prompt
- Analyst 6: open the full prompt

Point at

- Analyst 1: 195 characters, no evidence
- Analyst 6: the system prompt, the enforced answer schema, the inflation pre-read, the item ids
- The prompt hash: five repeats share it

Say

- If it is not on this page, the analyst does not get it.
- Same model. The only difference between analyst 1 and analyst 6 is this page.

LIVE DEMO

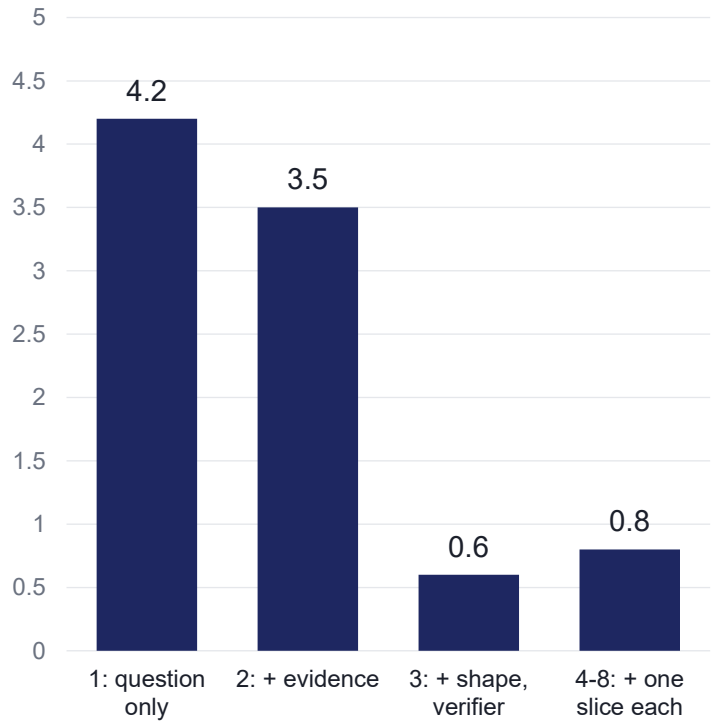
The ladder, four snapshots, five runs each

Dimension (pass mark)	Analyst 1: the question only	Analyst 2: + the evidence	Analyst 3: + answer shape + verifier	Analysts 4-8: + one slice each
S1 Repeatability (≤ 2 pp) max move of any probability across 5 identical runs	**3.4 / 4.5 / 4.9 / 4.1 pp, fail**	**3.3 / 2.3 / 4.8 / 3.4 pp, fail**	**0.6 / 0.5 / 0.5 / 0.8 bp of tilt, pass**	**0.5 / 0.8 / 1.2 / 0.8 bp of tilt, pass**
S2 Groundedness (≥ 95%) share of claims whose quote code found in the evidence	**0%, fail**	not measurable: no quotes	**100 / 100 / 100 / 99%, pass**	**100 / 97 / 100 / 100%, pass**
S3 Leakage (exactly 0) numbers in the answer that are not in the snapshot	**36 / 36 / 39 / 38, fail**	**0, pass**	**0, pass**	**0, pass**
What it concluded	Cut 25 bps at 57 to 67%: its memory of the Fed is stale	+25 bps at 50 to 57%: reads the evidence, overshoots the market	+7.0 / +6.6 / +6.4 / +5.8 bp hawkish	+4.4 / +4.1 / +4.4 / +4.0 bp hawkish; labour the lone non-hawk

Snapshots of 15, 24, 26 and 28 October 2026. Analysts 3 to 8 report S1 in basis points of tilt because they output a lean, not a distribution.

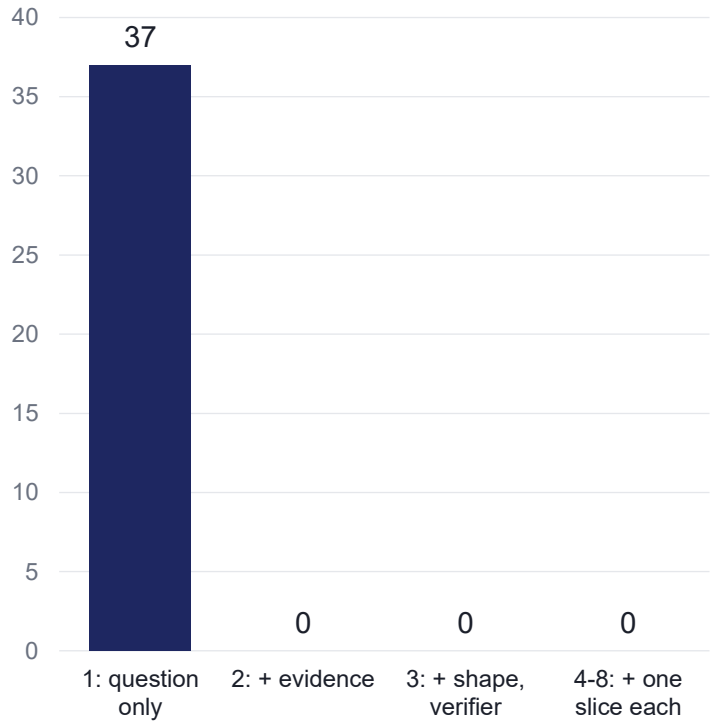
What each step bought

S1: wobble between identical runs (pp)



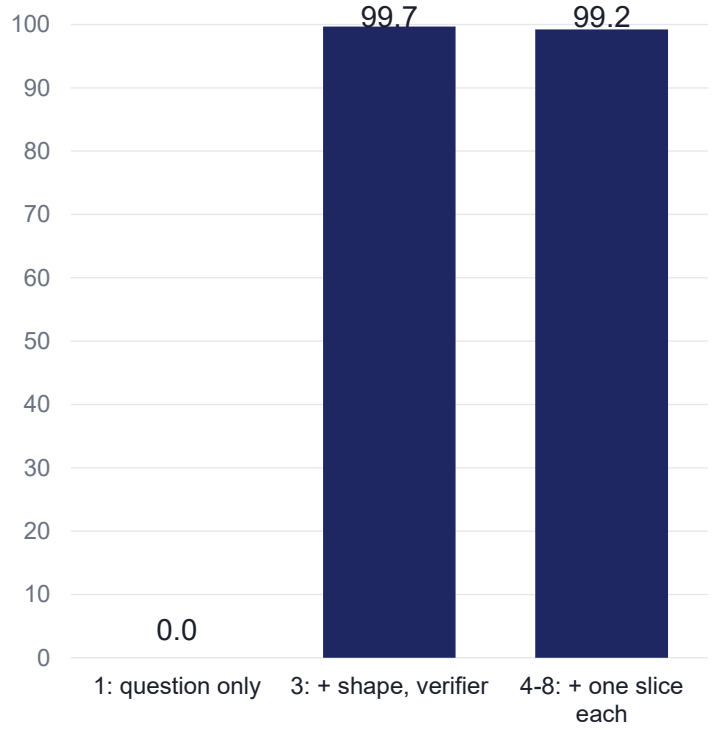
Largest standard deviation across five runs, mean of four snapshots. Pass mark 2 pp. Analysts 3 to 8: standard deviation of the lean, 1 bp = 1 pp.

S3: numbers from memory (count)



Numbers in the answer that exist nowhere in the frozen evidence, summed over five runs. Pass mark: exactly 0.

S2: claims with a verbatim quote (%)



Share of claims whose quote code found character for character in the item named. Pass mark 95%. Analyst 2 is not shown: it quotes nothing, so it cannot be measured.

The step from analyst 2 to analyst 3 is the harness: a fixed answer shape with a quote per claim, checked by code. It is the step that makes the model repeatable and checkable. Specialists keep both and add attribution.

What the specialists said on 26 October

Analyst	Reads	Lean (5 runs)	Verified
4 Communications	Both statements, minutes, transcript, speeches, tone pre-read	**+6.0 bp hawkish**	100%, 0 leaked
5 Minutes digest	The minutes only	**+6.6 bp hawkish**	100%, 0 leaked
6 Inflation	Inflation pre-read, projections, both statements	**+6.4 bp hawkish**	100%, 0 leaked
7 Labour market	Labour pre-read, projections, both statements	**–0.4 bp, neutral**	100%, 0 leaked
8 Financial conditions	Financial pre-read, base rates, both statements	**+3.4 bp hawkish**	100%, 0 leaked

One claim, as recorded

Communications analyst: the deterministic tone read scores the most recent policy communication as fully hawkish, with a unanimous hike vote and no dovish lexicon hits.

```
Quote: "score": 1.0   Item: signal.tone
Verified by code
```

The pre-read is our arithmetic over the two statements: 12 votes for, none against, eight hawkish words, no dovish ones. The analyst quoted it and code found the quote.

Live demo 4: what the analysts concluded

2 min

Open

- Analyst results view
- Analyst 1 (naive) card
- Analyst 4 (communications) card

Point at

- Analyst 1: cut 25 at 57%, numbers from memory flagged
- Analyst 4: the lean, grounded, leaked, watch-outs badges; one claim with its quote and the 'quote verified' mark
- Hover a badge for its definition

Say

- Every claim you see has a quote that code found character for character in the item named.
- The naive analyst still thinks the Fed is cutting. Same model, no evidence.

LIVE DEMO

The scorecard: eight of ten dimensions measured by code

S1 Repeatability measured by code	**S2** Groundedness measured by code	**S3** Leakage measured by code	**S4** Directional sanity measured by code	**S5** Noise invariance measured by code
S6 Paraphrase stability typed inputs only	**S7** Coherence measured by code	**S8** Reasoning quality not built: rubric grader	**S9** Auditability measured by code	**S10** Human-input discipline measured by code

Measured by code on the 26 October snapshot

- S4: 7 shocks, one input each, the pool must move the right way: 7 of 7
- S5: 5 irrelevant changes to the snapshot: 0.0 points moved
- S7: all 17 ledger entries coherent and attributable
- S9: entries under the current spec recompute exactly: 3 of 3

What it has already caught

- Four ledger entries stopped replaying after a model rule changed outside the hashed spec. Amendment #5 put every rule inside the hash; earlier entries are now labelled, not failed.
- A shock test was perturbing the wrong odds entry once two were on file. A red cell on the drift table pointed straight at it; fixed with a unit test the same hour.

Live demo 5: the scorecard and the drift table

2 min

Open

- Scorecard view
- The snapshot dropdown: pick 24 October, then 26 October
- Expand 'what was checked' under S4
- Scroll to the over-time table

Point at

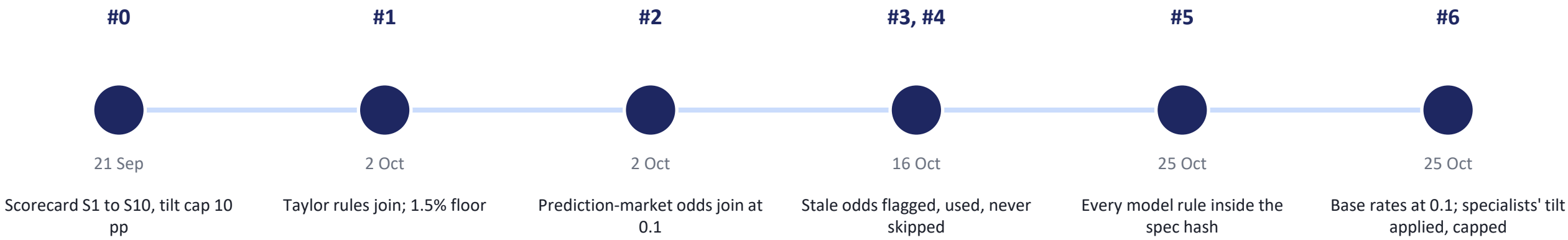
- The grid: red in the control columns, green in the judged columns
- S4's seven shocks with their deltas
- The over-time table: one row per snapshot, the marks never move

Say

- Analysts 1 and 2 are meant to fail. A red cell on the right would be a defect.
- This is the drift check. Two red cells have already pointed at defects and both are fixed.

LIVE DEMO

Governance: pre-register, amend in the open, replay



17

ledger entries, hash-chained

7

amendments, each with its reason

108

tests, spec hash pinned

0

numbers tuned on outcomes

Live demo 6: the ledger and replay

1 min

Open

- Ledger view
- Press Verify ledger

Point at

- 17 entries, each with its snapshot hash and code version
- The replay result: chain intact, entries under the current spec recompute exactly

Say

- Nothing in this ledger can be edited or deleted without the chain breaking.
- Any entry can be recomputed from its snapshot. That is what auditable means here.

LIVE DEMO

What this does not claim

No live record yet

The first decision this system is scored on is 28 October 2026. Everything so far is process evidence, which is the point, but calibration needs a record.

Weights are judgement

Market 0.6, Taylor 0.2, odds 0.1, base rates 0.1, the floor and the tilt mapping were set by reasoning and logged, never fitted. Candidate sets will be pre-registered and scored live.

The fitted model is outside

A first ordered probit failed its out-of-sample gate and was reported, not tuned. Version 2 beats the base rate but carries a contamination caveat, so it stays at weight 0.

Small, unbalanced history

About 130 decisions with full features, roughly 250 in all; 68% are holds and cuts cluster in three episodes. That is why there is no gradient-boosted anything.

Not every dimension yet

Reasoning quality (S8) needs a rubric grader checked against hand grades; the reworded-prompt test for the analysts is not built.

The market is the anchor

At thirty days futures call most decisions. The claim is auditability and discipline, not a better hit rate at this horizon.

Follow-ups, in order of value

1 Score live decisions: built this week

Once the overnight-rate row is dated after the meeting, one click scores every forecast of that meeting, each track, each model and two naive baselines, with log score and Brier, into an append-only file. The first record lands the morning after 28 October. Every weight decision waits on this record.

2 Longer windows, same chain

The December meeting and the year-end rate. The market is less certain there (Polymarket prices a December hike at 76%), so evidence reading has more to add. The chain needs only a different target.

3 An events agent for regime shifts

1987, 1998, 2008, 2020: the Fed cut because of financial stress, not the data. A watch-out agent that scans the last three months for bank failures, funding stress and market dislocations, and raises a typed watch-out every analyst must answer.

4 Pre-registered weight sets

Three or four candidate weight sets written down now, all scored on each live decision; the live set changes only by amendment on that record.

5 Reasoning quality, S8

Twenty traces hand-graded on four questions; a model grader trusted only once it agrees with those grades.

6 One more step and a second model family

A hawk, a dove and a synthesiser on the ladder; a second model family as a cross-check that the harness, not the model, is doing the work.

Three things to take away

- The harness is the product. Same model, same evidence, same day: naive it wobbles and leaks; harnessed it is stable, quoted and checked.
- Judge the process. Eight decisions a year cannot prove skill; a pre-registered scorecard measured on every snapshot can show discipline and catch drift.
- The ask: scrutinise the scorecard, suggest the next window, and tell us which other decisions deserve the same chain.